CAP182 Capstone Project

SS3, Part A: Peer Review

A critique of the motivation, problem statement, model choices and statistical analysis of a peer capstone project

Reviewer

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Module	CAP182 Capstone Project
Assessment	SS3 Part A: Peer Review

Project under review

Student reviewed	Lavani Chauke
Student number	26304796
Project title	Beyond the Funnel: behavioural segmentation of STADIO 1's free app and fantasy audience
Client case	STADIO 1 (global motorsport and media)
GitHub repository	https://github.com/Lavani444/SS1-Capstone-project-Lavani-Chauke-26304796

Reviewed student details

Name: Lavani Chauke

Student number: 26304796

GitHub: https://github.com/Lavani444/SS1-Capstone-project-Lavani-Chauke-26304796

The project reviewed is “Beyond the Funnel”, a behavioural segmentation of STADIO 1's free app and fantasy audience and its link to churn. I went through and reviewed the SS1 documents which include motivation, problem statement, data request and RAAIDD log and the SS2 repository: the two models (K-Means and Logistic Regression), the output files and metrics, the literature review and the recommendations report.

Motivation

What works: The motivation is concrete and accurate. The figures it uses (18 million free users and 3.4 million subscribers, free-to-paid conversion down from 4.9% to 3.6%, subscriber churn up from 19% to 26%, app engagement per race down from 38 to 31 minutes, and sponsor renewal down from 88% to 81%) match the STADIO 1 company pack. The board's priority is to know every fan as an individual, and the motivation ties the problem to this priority by naming those who are affected and showing the scale (one percentage point of conversion across 18 million users is 180,000 subscribers). It is also honest that segmentation will not fix conversion or churn on its own but gives decision makers an evidence base.

Concerns: 1. The suggestion that many free users may be content with the free product goes beyond the pack, which only says that conversion falls as the free product improves.

2. Some wording drifts from the source: the pack says newer fans churn faster, not "younger and more chaotic" fans, and "fan-to-fan pillar" should be direct-to-fan.

Suggestion: Even though the RAAIDD log records why this problem was chosen (decision D1), the Motivation itself does not. A short comparison of two or three candidate problems on impact, data availability and feasibility, with a brief reason why segmentation was chosen over the alternatives, would make the Motivation stronger.

Problem statement

What works: The statement is anchored in measurable symptoms (free-to-paid conversion falling from 4.9% to 3.6% and annual streaming churn rising from 19% to 26%), and the scope is clear from the RAAIDD log. The Part C data request is well aligned to it: pseudonymised fan-level data, UTC timestamps, identity match method and confidence columns, a request that the client document its rules for “Lapsed” fans and cancellations, and a justified exclusion of social data.

Concerns: 1. The statement is two sentences and closes with “It’s also unclear if different fan behavioural groups can be distinguished…”. That is a discovery question without a success criterion: nothing says what would count as distinguishable (for example stable clusters, or clusters that differ significantly in conversion or churn).

2. It combines three issues (an undifferentiated audience, falling conversion and rising churn), but the SS2 evidence tests only churn, because the proxy dataset has no conversion outcome.

3. The objective, which names the users (the digital and commercial teams) and the use (targeting offers and communications), appears as an unlabelled paragraph at the end of the README, after the RAAIDD log. The Problem Statement itself does not say who acts on the result or how, so the reader has to join the two.

Suggestion: Merge the problem and objective into one statement with a population, an outcome and a test. For example: identify behavioural fan segments that are stable and that differ significantly, with a meaningful effect size and after multiple-test correction, in free-to-paid conversion and in streaming churn, so that the digital and commercial teams can target offers by segment. Choose conversion or churn as the primary outcome and treat the other as secondary.

Model choices

What works: The pipeline is cohesive. K-Means generates segments independently of the churn label (decision D4), and Logistic Regression then tests whether the segments

enhance predictive performance using a baseline model and a with-segment model with identical settings. The practice is careful: scaling is fitted on the training split only, k is chosen from silhouette within 3 to 6 and cross-checked with Davies-Bouldin and Calinski-Harabasz, stability is tested over 100 resamples (ARI 0.98), class weights handle the 27% churn rate, and the model is tuned on PR-AUC instead of accuracy. The documentation states its limits (K-Means assumes compact clusters, a linear model misses interactions, and gradient boosting is left as a literature comparison). The literature review is relevant and the choices follow it.

Concerns: Tenure and contract length are among the strongest churn drivers in the model (tenure has the largest coefficient in coefficients.csv), and they are also two of the five clustering features. Indeed, all five clustering features are also inputs to the baseline Logistic Regression. No churn label leaks into the segments (decision D4), but the segments are built from variables the baseline already uses. A segment–churn association is therefore partly expected, and the finding that segments add almost nothing (PR-AUC +0.001) is unsurprising. A stronger test would cluster on behavioural features only, or compare segments against the baseline on a measure the baseline cannot already see.

Suggestion: First, check that the segment–churn pattern survives when the clustering excludes tenure and contract length, or show that the segments add value beyond the baseline model using features the baseline cannot already see. Second, compare at least one alternative clustering method (for example a Gaussian mixture or hierarchical clustering), as the report already proposes. Optionally, add one tree-based classifier to test whether the linear model leaves useful signal unused.

Statistical analysis

What works: The analysis is rigorous and mostly well suited to the questions. Test metrics for the baseline model come with bootstrap intervals (ROC-AUC 0.834, 95% CI 0.811 to 0.857), and PR-AUC is compared with the no-skill level of 0.266. The decision threshold was chosen on out-of-fold training predictions and not on the test set. Pairwise segment churn tests use a Holm correction, effect sizes (Cramér’s V 0.39 to 0.42) are

reported next to chi-square p-values, and model comparisons use a paired bootstrap. The report also states a negative result: adding the segment did not improve the model (PR-AUC change +0.001, p = 0.44). It reports calibration and blind spots too: mean predicted risk of 0.42 against 0.27 observed, which is consistent with class-weighted training, and recall of only 30% in segment S2 and 9% in S4.

Concerns: 1. McNemar's test for baseline against with-segment gives p = 0.004 (9 against 27 disagreements), but the paired bootstrap gives p = 0.42 for ROC-AUC and 0.44 for PR-AUC. The two models use different thresholds (0.55 and 0.58), so McNemar is probably detecting the threshold shift, not a better model, and the McNemar result appears only in the output file and is not discussed in the report.

2. Everything rests on one 80/20 split with 374 test churners. Cross-validation on the training data and bootstrap intervals help, but they cannot show how far the results move between splits, and some cells are very small: segment S0 has only 1 churner among 189 test customers, and S1 against S3 (Holm p = 0.20) and S2 against S4 (p = 0.38) are not significant on test, so only 8 of 10 segment pairs differ there.

Suggestion: Report McNemar's test at a common threshold, or at the same alert rate (his gain curves already allow this), or drop it, so that it does not contradict the bootstrap. Use repeated stratified splits or repeated cross-validation to show how much the results move. As the report proposes, apply Platt or isotonic calibration before presenting probabilities to business users, and choose the threshold by cost once cost data exists.